

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:	K.Karthik	Reg. No.:	192525040
Section / Batch:		Date:	15/07/2026

Code of Conduct

I _____K.Karthik(192525040)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Karthik

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic		Focus	Q Nos	Marks
Data Structure		Real-Time Applications	1	100
GRAND TOTAL:			100 Marks	

Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Mark)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks)

(OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	20	Demonstrates deep and accurate understanding of data structure with insights and connections	Shows clear understanding with minor gaps	Basic understanding; some misconceptions present	Limited or incorrect understanding
Content Organization	30	Logical, well-structured, smooth flow; strong introduction and conclusion	Mostly organized with clear flow	Some organization but lacks clarity or transitions	Disorganized, difficult to follow
Technical Depth	20	Includes algorithms, complexity analysis, and advanced insights	Includes algorithm and basic complexity analysis	Limited technical details; missing depth	Minimal or no technical content
PowerPoint Slides	20	Excellent design, clear visuals, enhances understanding	Good slides with minor issues	Basic slides; somewhat cluttered or unclear	Poor slides or no visual support
Communication Skills	10	Clear, confident, engaging delivery; excellent articulation	Clear delivery with minor hesitation	Some hesitation; unclear in parts	Poor communication; difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0305-Data Structure

Metro Rail Network – Reachable Stations

Guided by:
Dr.Uma Priyadarshini

Presented by:
K.Karthik
192525040



ABSTRACT



- Metro rail networks consist of stations connected by routes.
- The system must determine all stations reachable from a given source station.
- Choosing an appropriate data structure improves route searching and navigation.
- A Graph is the most suitable data structure because it efficiently represents stations and their connections.



INTRODUCTION

- A metro network consists of multiple stations connected by railway tracks.
- Users need to find reachable stations and travel paths.
- The data structure should efficiently represent stations and routes.
- Graphs are widely used in navigation and transportation systems.



PROBLEM STATEMENT

Problem:

A metro rail network maps stations and routes to determine reachable stations from a source station.

Requirement:

Recommend the most suitable data structure based on flexibility, efficiency, and route traversal.



Recommended Data Structure – Graph



Why Graph?

- Stations are represented as Vertices (Nodes).
- Routes between stations are represented as Edges.
- Easily represents complex metro networks.
- Supports Breadth-First Search (BFS) to find all reachable stations.
- Efficient for route planning and shortest path algorithms.



Advantages of

Using Graph

Benefits

- Efficiently represents interconnected stations.
- Easy to add or remove stations and routes.
- Supports route planning algorithms.
- Finds all reachable stations using BFS or DFS.



Real-World Applications



Applications of Graphs in Transportation

- Metro Rail Networks
- Google Maps Navigation
- Railway Reservation Systems
- Airline Route Planning
- Road Traffic Management
- GPS Navigation Systems
- Delivery Route Optimization



CONCLUSION

- A Graph is the best data structure for representing a metro rail network.
- It efficiently models stations and routes.
- BFS or DFS can determine all reachable stations from a source.
- Therefore, graphs provide an effective and scalable solution for metro route management.



Engineering to Revolve

SIMATS
ENGINEERING



SIMATS^{II}

SIMATS Institute of Medical and Technical Sciences
Recognized As Deemed to be University under Section 3 of UGC Act 1956

0305-Data Structure Metronetwork – Reachable Stations

Guided by
Dr. Uma

Presented by
K.Karthik
192825040